Class	Adm No

Candidate Name: ______________________________

H

millennia institute

2017 Preliminary Examination 2
Pre-University 3

H2 Biology **9648/02**

Paper 2 Core Paper **13 September 2017**

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer any **one** question.

The use of an approved scientific calculator is expected, where appropriate. You will lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

For Examiner's Use

Section A

Answer **all** questions in this section.

1. Fig. 1.1 shows a series of micrographs of animal cells undergoing cell and nuclear division.

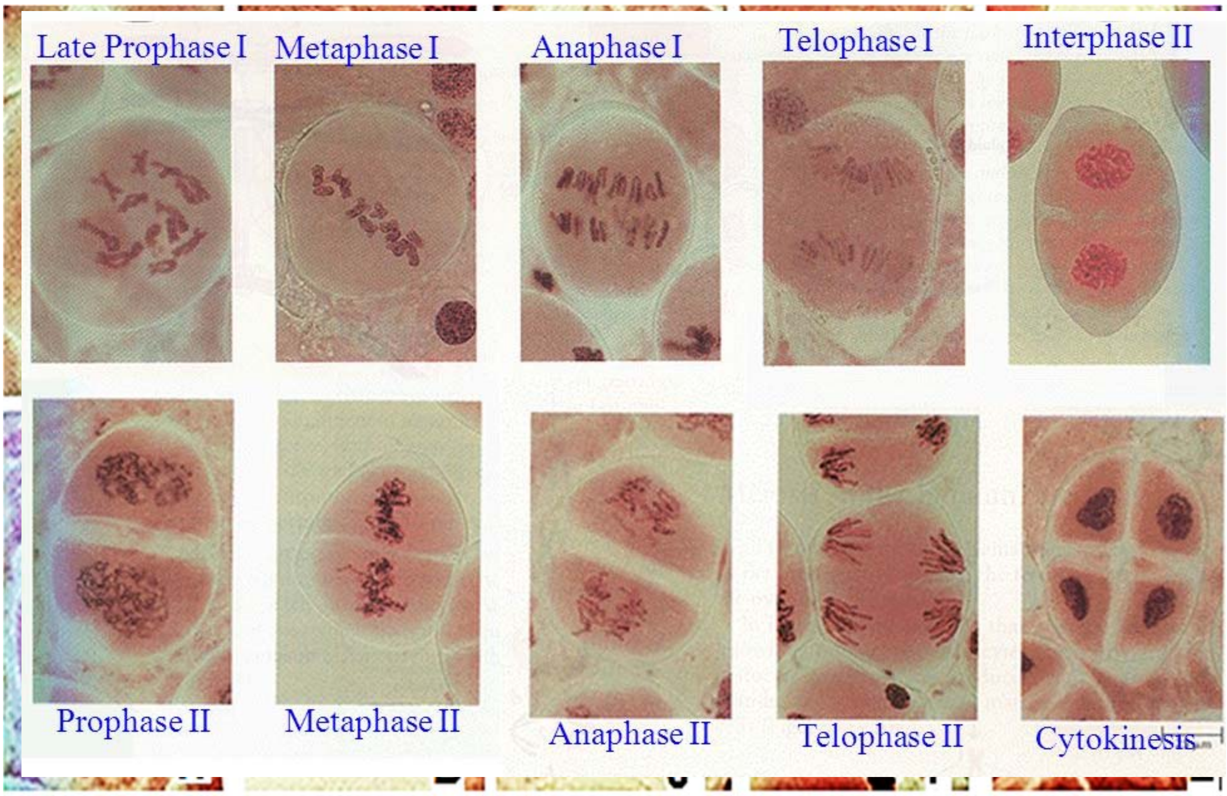


Fig. 1.1

(a) Arrange the letters in Fig. 1.1 in a correct sequence to show the events occurring in the cell and nuclear division process.

A, F, C, D, G, H, B, J, I, E.

; 5 letters in correct sequence for 1 mark, max is 2 mark

.....**[2]**

(b) Describe the processes occurring in **A**.

....

- Pairing of homologous chromosomes to form bivalents / synapsis occurs;
- Crossing-over of alleles at chiasmata;
- between non-sister chromatids of the homologous chromosomes;
- creates new combination of alleles and giving rise to genetic variation in the gametes and hence offspring;
- The chromatids / euchromatin condense to form chromosomes;
- The nucleolus and nuclear envelope disintegrate;
- Centrioles migrated to poles and spindle forms;

....

....

...**[2]**

b

For Examiner's Use

(c) State one similarity and one difference between process **C** and **J**.

- Similarity 1: Chromatids are separated;
- Similarity 2: Shortening of the microtubules / spindle fibres;
- Similarity 3: Genetic material migrate towards opposite poles of the cell / towards centrioles;

- Difference 1: The number of chromosomes in anaphase II is half of that in anaphase I;
- Difference 2: The centromeres in process C do not divide but the centromeres in process J divides;
- Difference 3: Homologous chromosomes separate during C but chromatids separate during J;

2]

(d) Explain the significance of process **H**.

- Prevent the doubling of chromosome number in the organism;
- Reduction of the number of chromosome number by half for gametes;
- So that when the gametes fuse, the original chromosome number of the cell will be restored;

R! Haploid

..[2]

(e) Suggest how the cell and nuclear division process would be affected if centromeric DNA is deleted from a chromosome.

- Chromatids are not held/joined together;
- Kinetochores cannot form on chromosome;
- Spindle fibres cannot attach/bind to chromosomes;
- chromosomes cannot align along equatorial plane / metaphase plate during metaphase;
- Ref. to random movement of sister chromatids / chromatids not separated to opposite poles *(idea of opposite poles is important)* ;
- Daughter cells will not have complete set of chromosomes/ have extra or less chromosomes / will not be genetically identical to parent cell / ref. to aneuploidy;
- mitosis may not proceed past metaphase / anaphase cannot occur;
- Non-disjunction can occur;

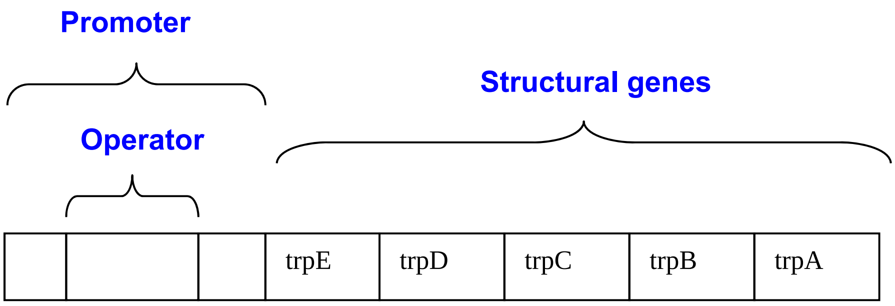
[2]

l 10]

2. In bacteria, the production of the amino acid tryptophan is catalyzed by five specific enzymes (simply named as **E, D, C, B** and **A** in this question) encoded by specific genes *trpE, trpD, trpC, trpB and trpA*. The *trp* operon is transcriptionally regulated by a repressor protein, (named **R** in this question), encoded by the *trpR* gene. Expression of the *trpE, trpD, trpC, trpB* and *trpA* genes is controlled by a promoter region and an operator region.

When levels of tryptophan are high, tryptophan binds to the repressor protein, **R**. The tryptophan-repressor protein complex binds to the operator region and prevents expression of the *trpE, trpD, trpC, trpB* and *trpA* genes.

(a) Draw a simple diagram to show the *trp* operon. **[1]**



Reject: if operator not within promoter
Reject: if the structural genes are not in order of sequence; all or none

(b) Explain why it is useful for a bacterial cell to decrease expression of the *trp* genes when tryptophan is present.

- Trp genes code for enzymes (involved in/necessary for) <u>(anabolism/synthesis) of tryptophan</u>;
- Decreased expression helps to <u>conserve resources</u> that could be diverted for other uses; accept idea of <u>preventing wastage of resource</u>;

Reject: ONLY mention of energy and not resources

.... **[2]**

For Examiner's Use

Table 2.1 below indicates the activity levels of the functional enzymes E, D, C, B and A in wild type bacterial cells in the presence and absence of tryptophan (Trp).

Table. 2.1

	Activity level of enzymes/units	
Enzyme	**Trp absent**	**Trp present**
E	700	0
D	700	0
C	700	0
B	700	0
A	700	0

Researchers have managed to obtain several bacterial mutants. Each mutant is the result of a single base-pair substitution in a single component of the *trp* operon. The activity level of functional enzymes E, D, C, B and A in the bacterial cells having these individual mutations is shown in Table 2.2.

Table. 2.2

	Activity level of enzymes/units					
	Mutant 1		**Mutant 2**		**Mutant 3**	
Enzymes	**Trp absent**	**Trp present**	**Trp absent**	**Trp present**	**Trp absent**	**Trp present**
E	700	700	700	0	0	0
D	700	700	0	0	0	0
C	700	700	700	0	0	0
B	700	700	700	0	0	0
A	700	700	700	0	0	0

(c) With reference to Table 2.1 and Table 2.2, identify the mutant bacteria that has a phenotype that is consistent with a loss-of-function mutation in the *trpR* gene and explain your choice.

- Mutant 1;
- The normal / wild type *trpR* activity is repression of the *trp* operon / codes for repressor protein (e.g. zero enzyme activity in the presence of tryptophan);
- A loss of the repression / lack of repressor will lead to constitutive expression of the *trp* genes (700 units of enzyme activity even in the presence of tryptophan);

……………………………………………………………………………………**[2]**

(d) Account for the phenotype of mutant 3 assuming that it experienced a loss-of-function mutation.

- A loss of function of the promoter;
- 0 units of enzyme activity regardless whether tryptophan is absent or present;
- Transcription factors are not able to the bind to the promoter to form transcription initiation complex;
- RNA polymerase is unable to bind to the promoter to initiate transcription and thus block / prevent expression of the *trp* operon;

Reject: operator as a loss of function in that component would not allow the repressor to bind, causing expression of the *trp* genes.
Reject *trpR* loss-of-function mutation since there would be no functional repressor and the phenotype would be the same as mutant 1.

…….

………

…….

….[2]

(e) If the phenotype of mutant 3 is caused by a mutation in the *trpR* gene, explain how this mutation would affect the structure and function of the repressor protein.

- Gain of function mutation which causes a change in amino acid / Missense mutation in the *trpR* gene;
- which causes the protein to fold into the same conformation as the active repressor / trp-bound wild type R protein; (only awarded if it leads to correctly stated altered repressor function as stated below);
- trp repressor is always existing in an active conformation / constitutively active;
- Permanently bound to operator / binds to operator even in absence of tryptophan;
- Prevent binding of RNA polymerase to the promoter and inhibit transcription of genes;

…….

…….

…….

….[2]

[Total: 9]

For Examiner's Use

3. Researchers are constantly investigating the effects of limiting factors on the rate of photosynthesis on various plants. Fig. 3.1 show how three main limiting factors, carbon dioxide concentration, light intensity and temperature can affect the rate of photosynthesis in cactus plants.

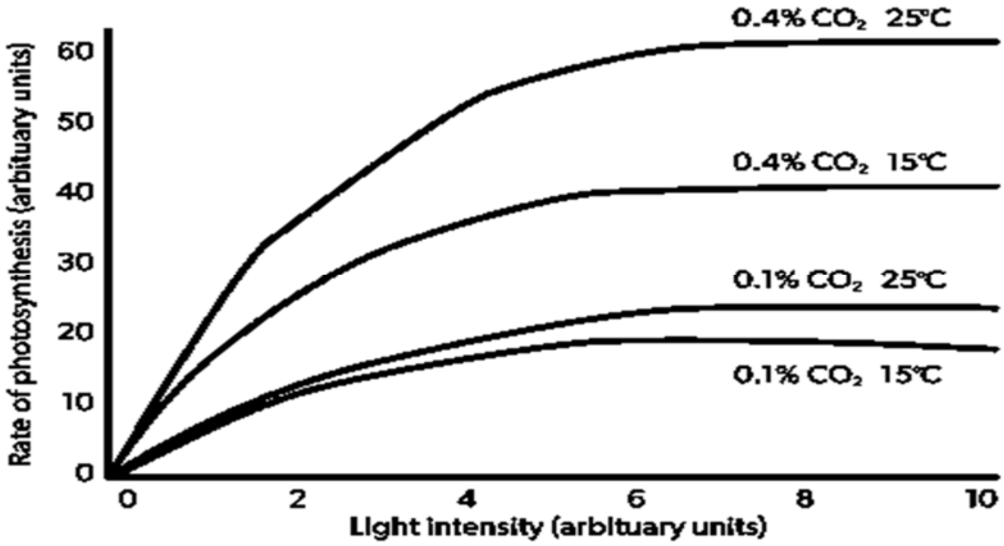


Fig. 3.1

(a) Define the term 'limiting factor'.

- Rate of photosynthesis is limited by the slowest reaction in the series and when the rate of photosynthesis is limited by more than one factors;
- the rate is limited by the factor which is in the shortest supply;
- This factor determines the rate of reaction, ie. when this factor is increased in concentration, the rate increases too;

[1]

(i) explain the effect of light intensity on rate of photosynthesis.

- As light intensity increases from 0 to 6 arbitrary units, rate of oxygen production increases / rate of photosynthesis increases proportionally with increasing light intensity;
- At low light intensities, light intensity is the limiting factor;
- Light is needed for photolysis of water to provide electrons for photophosphorylation;
- Light directly affects light-dependent stage / non-cyclic + cyclic photophosphorylation of photosynthesis which is responsible for synthesis of ATP and NADPH;
- More ATP and NADPH produced per unit time would lead to increase rate of Calvin cycle;
- More G3P converted to glucose per unit time;

- As light intensity increases above 6 arbitrary units, rate of oxygen production remains constant at the maximum rate;
- Light intensity is no longer a limiting factor, other environmental factors like carbon dioxide concentration or temperature or other factors may be limiting;
- When the light saturation point being reached, any further increase in light intensity will have no effect on the rate of light dependent reaction / rate of oxygen production;

[5]

For Examiner's Use

(ii) justify if carbon dioxide concentration or temperature is a greater limiting

- Carbon dioxide;
- A 0.3% increase in carbon dioxide increases rate of photosynthesis by 25au at 15°C / 40au at 25°C but a 10°C increase in temperature increases rate of photosynthesis by 5au at 0.1% CO2 concentration / 20au at 0.4% CO2 concentration;
- Increasing the temperature to 25°C will increase rate of photosynthesis to 20au but increasing the carbon dioxide concentration to 0.4% will increase the rate of photosynthesis to 60au;
- Carbon dioxide is needed for carbon fixation during Calvin cycle when combining with RuBP;
- Temperature affects mainly light-independent stage and rate of photosynthesis doubles for each rise of 10°C until optimum temperature is reached;
- However, rate of photosynthesis decreases at higher temperature as enzymes start to denature;

[3]

(c) Suggest why water is not considered a limiting factor for the rate of photosynthesis.

- Amount of water needed for photosynthesis is very little;
- When plants do not have enough water, stomata on leaves will close and limit amount of carbon dioxide entering the plant;
- Plants wilt when there is insufficient water;
- AVP;

[1]

Fig. 3.2 illustrates a graph showing how varying light intensity affects the net carbon dioxide uptake and release in sun and shade plants.

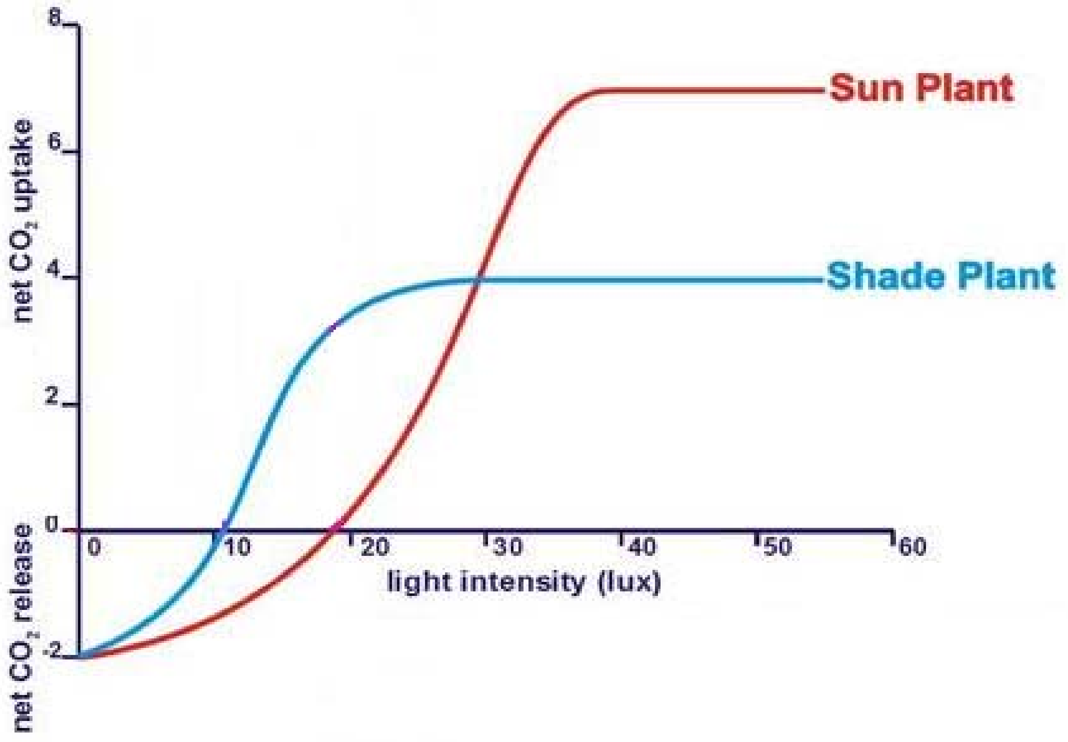


Fig. 3.2

For Examiner's Use

(d) With reference to Fig. 3.2,

(i) state the compensation points of sun and shade plants.

- Sun plants: 19 lux;
- Shade plants: 11 lux;

....[1]

(ii) account for the graph differences between sun and shade plants.

- Compensation point is lower for shade plants by 9 lux as compared to sun plants;
- Net carbon dioxide uptake is higher for sun plants than shade plants by 2.5 – 3 au;
- Shade plants reaches maximum net carbon dioxide uptake of 4 au at 28 lux whereas sun plants reaches maximum net carbon dioxide uptake of 7 au at 37 lux;

Max 2 marks

- Kreb cycle occurs more at low light intensities from 0 – 20 lux for sun plants;
- Carbon fixation during Calvin cycle occurs more at higher light intensities from 19 lux and above for sun plants;

OR

- Shade plants have more chloroplast;
- Can absorb light even at low light intensities;
- and light-dependent reactions can still occur at low-light intensities;
- Rate of photosynthesis is higher than rate of respiration at low light intensities from 10 lux and above;

.....[4]

otal: 15]

2012 MJC Prelim 2 Q6

For Examiner's Use

4. A study was conducted to study the inheritance of coat colour in mice, in which one of the allele is also known to affect normal embryonic development. A cross between agouti mouse (with agouti coat colour) and yellow mouse (with yellow coat colour) resulted in half of the F1 progeny being agouti mice and the other half being yellow. Mating of F1 yellow mice resulted in the following F2 generation.

Agouti mice 98
Yellow mice 202

(a) Using the symbols **A** and **a** for the two alleles involved, draw a genetic diagram in the space below to show how the F1 cross resulted in the F2 progeny.

Key: **A represents the dominant allele for yellow coat colour.**
a represents the recessive allele for agouti coat colour.

F1 phenotypes	Yellow x Yellow
F1 genotypes	Aa x Aa
F1 gametes	A a x A a
F2 genotypes	AA Aa Aa aa
F2 phenotypic ratio	202 Yellow : 98 Agouti 2 Yellow : 1 Agouti
	AA died

[3]

In another experiment involving deer mouse, pure breeding pink-eyed mice with wild-type fur was crossed with pure breeding dark-eyed albino mice. The resulting progeny all had wild-type fur and dark eyes. These F1 mice were then crossed with pink-eyed albino mice. The results are shown in Table 4.1. It was difficult to distinguish between mice that are dark-eyed albino and pink-eyed albino, so these two phenotypes were counted together.

Table 4.1

Phenotype	**Number of progeny**
Wild-type fur, dark-eyed (recombinant)	12
Wild-type fur, pink-eyed (parental)	62
Albino, dark-eyed (parental) Albino, pink-eyed (recombinant)	78
Total	152

(b) In the blank space below, calculate the chi-square value. **[2]**

Phenotype	Observed (O)	Expected (E)	$(O-E)^2$ / E
Wild-type fur, dark-eyed	12	38	17.7
Wild-type fur, pink-eyed	62	38	15.2
Albino, dark-eyed Albino, pink-eyed	78	76	0.05
		Chi Square Value	32.95

Table 4.2 shows a portion of the chi-square table.

Table 4.2

distribution of X^2			
number of degrees of freedom (v)	**probability**		
	0.1	**0.05**	**0.01**
1	2.71	3.84	6.64
2	4.60	5.99	9.21
3	6.25	7.82	11.34
4	7.78	9.49	13.28

For Examiner's Use

(c) Using the values in Table 4.2, draw appropriate conclusions as to whether the results of the cross followed the expected ratio you have predicted in **(b)**.

- P(difference occurring by chance) <0.05 for df of 2;
- Critical value < calculated chi-square value of 32.95;
- difference between observed and expected results is significant / not due to chance;
- Does not conform to the expected phenotypic ratio of 1:1:1:1 / 1:1:2;
- Hence the two genes do not segregate independently / the two genes are linked;

[2]

(d) Using **E/e** to represent alleles for eye colour and **A/a** to represent alleles for fur coat colour, explain the result of the F1 cross in the deer mouse experiment using a genetic diagram.

Key: E represents the dominant allele for dark-eyed.
e represents the recessive allele for pink-eyed.
A represents the dominant allele for wild-type fur.
a represents the recessive allele for albino

F1 Cross:

Phenotype of parents: dark-eyed, wild type fur x pink-eyed, albino

Genotype of parents: e A / E a x e a / e a

Gametes produced (n): (e A) (E a) (e a) (E A) x (e a)

Genotype of offspring: e A / e a, E a / e a, e a / e a, E A / e a

Phenotype of offspring: pinked-eyed wild-type fur, dark-eyed albino, pink-eyed albino, dark-eyed wild-type fur

Parental type   Recombinant type

Phenotypic ratio: 62 : 78 : 12

[4]

[Total: 11]

5. *Bungarus multicinctus*, also known as the Taiwanese Krait, is a species of venomous snake endemic to Asia and is predominantly found in forests from Taiwan to Southeast Asia. In order to better understand the venomous snake's physiological pathways, researchers have been conducting extensive research on the mechanism of action of Taiwanese Krait venom which consists primarily of neurotoxins.

Fig. 5.1 shows a diagram of the kappa-bungarotoxin, a neurotoxin found in the venom of the Taiwanese Krait. Kappa-bungarotoxin is a highly stable protein molecule that is capable of withstanding harsh chemical reactions.

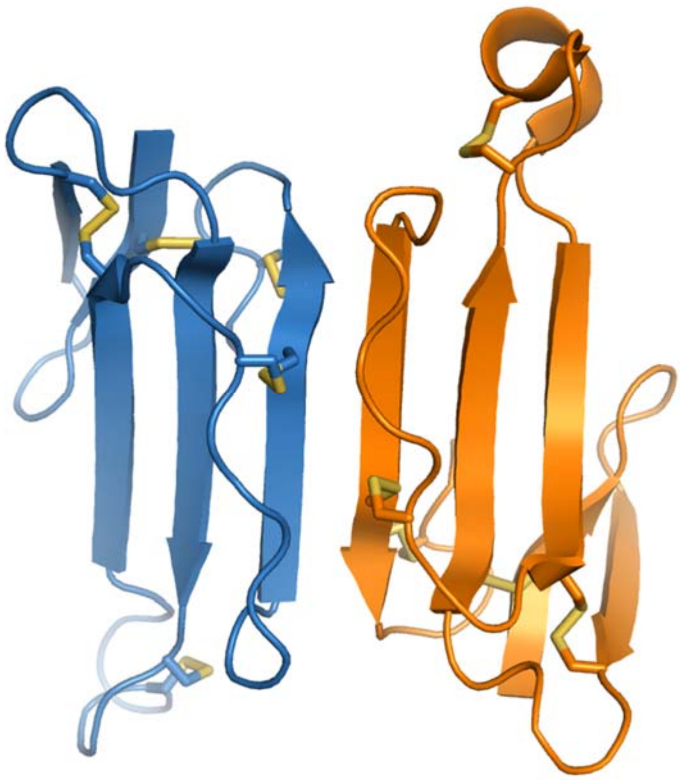


Fig. 5.1

(a) Explain how the protein structure of kappa-bungarotoxin is maintained.

- Secondary structures include beta-pleated sheets held together by intramolecular hydrogen bonds along the polypeptide backbone;
- Tertiary structure: 3D configuration held together by bonds such as hydrophobic interactions, disulphide bridges, ionic bonds and hydrogen bonds between R-groups;
- Quaternary structure: 2 subunits are held together by hydrophobic interactions / covalent bonds;

[3]

For Examiner's Use

People bitten by *Bungarus multicinctus* suffer from neuromuscular paralysis and respiratory failure. Research shows that the venom causes serious health complications due to the effects of kappa-bungarotoxin at the neuromuscular junctions at the muscle cells of the lungs as shown in Fig. 5.2.

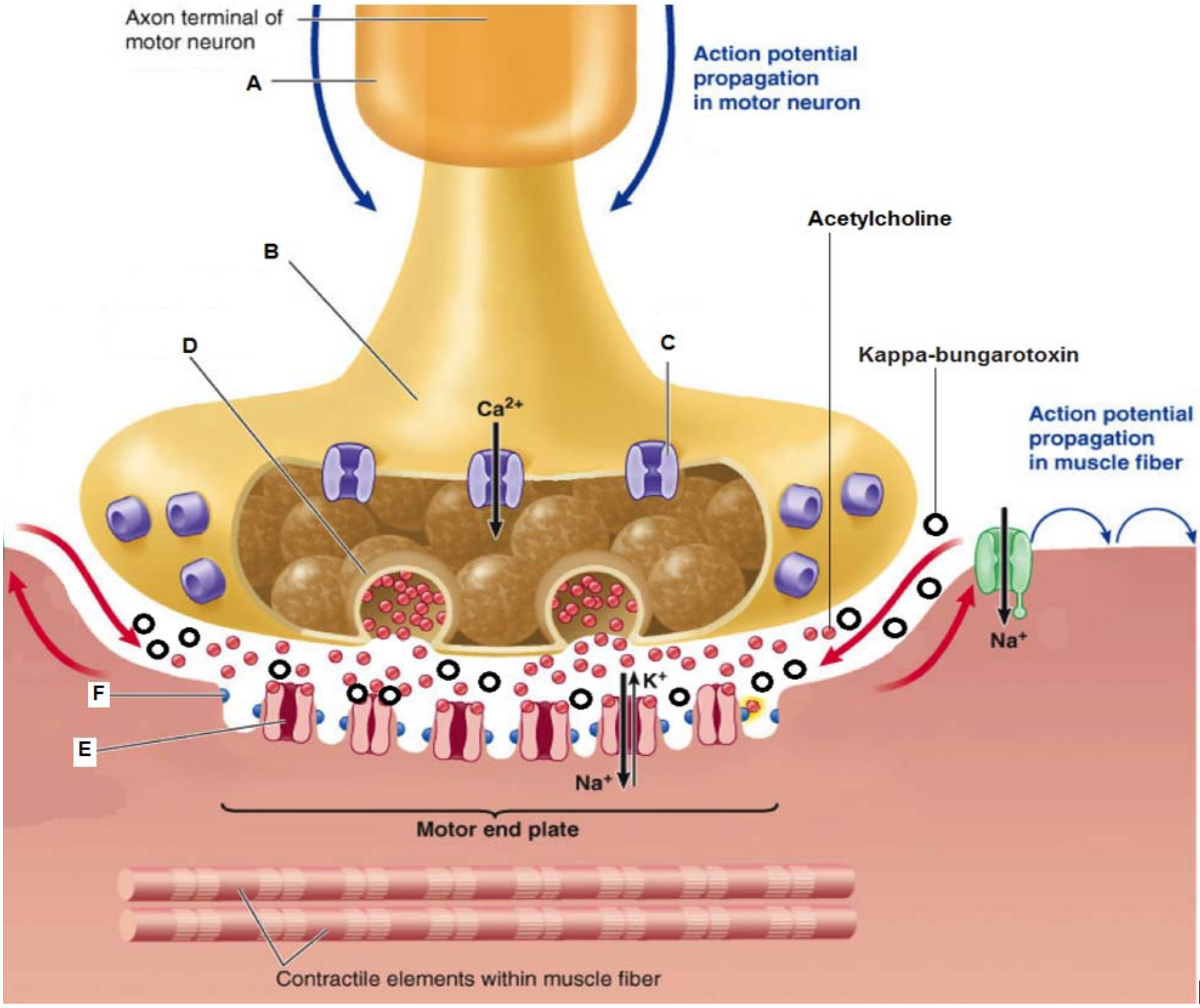


Fig. 5.2

(b) Name the parts of the neuromuscular junction shown in Fig 5.2 labelled **A**, **B**, **C, D** and **E**.

A: ......

B: ......

C: ......

D: ......

A: Myelin sheath;
B: Presynaptic knob;
C: Voltage-gated calcium ion channels;
D: Secretory / Synaptic vesicles (containing acetylcholine);
E: Cholinergic receptors / Post-synaptic receptors / Ligand-gated ion channel;

2 correct – 1 mark
5 correct – 2 marks

E: ....

[2]

In Fig. 5.2, **F** is an enzyme important in synaptic signalling.

(c) Describe the function of **F**.

- Aceylcholinesterase breaks down acetylcholine into acetyl group and choline;
- to prevent over-generation of action potentials at the post-synaptic terminal;

[1]

(d) Explain how the transmission of nervous impulse shown in Fig. 5.2 differs from electrical transmission of action potentials.

	Transmission of action potential along a neurone		Synaptic transmission
1	Electrical in nature.	1	Chemical in nature.
2	Action potential travels along one neurone.	2	Involves two adjacent neurones.
3	Calcium ion is not required.	3	Calcium ions are required for triggering the release of neurotransmitter into the synaptic cleft.
4	Only excitatory.	4	Synapses can be excitatory or inhibitory.
5	Depolarisation is due to an influx of sodium ions, resulting from an increase in the permeability of axon membrane to sodium ions.	5	In an excitatory synapse, the post-synaptic membrane becomes more permeable to both sodium ions and potassium ions. Depolarisation (or EPSP) is due to greater inward diffusion of sodium ions into the post-synaptic neurone than outward diffusion of potassium ions.
6	Action potential occurs immediately in response to a stimulus of adequate strength.	6	There is a synaptic delay. This is due to time taken for the release of the neurotransmitter from synaptic vesicles, and for the diffusion of the neurotransmitter across the synaptic cleft.
7	Action potential is an all-or-none response.	7	Several EPSP can be summated to reach the threshold potential.
8	Recovery is due to repolarisation (influx of sodium ions stops and efflux of potassium ions) and sodium-potassium pump.	8	Recovery requires acetylcholinesterase, which hydrolyses acetylcholine.
9	An absolute refractory period and a relative refractory period exist.	9	There is no refractory period but synapse can be fatigued due to a shortage of neurotransmitter.
10	Speed of transmission is affected by the diameter of axon, presence or absence of myelin sheath and temperature.	10	Synaptic transmission can be affected by drugs, eg curare. Curare binds to the receptor sites for acetylcholine.
11	Bi-directional; depolarisation waves sweep along both sides of stimulus.	11	Synaptic transmission is always unidirectional, from the pre-synaptic membrane to the post-synaptic membrane.
12	Information is transmitted over a distance of one metre.	12	Information is transmitted over a distance of 20 nm.

[2]

(e) Explain how kappa-bungarotoxin causes respiratory failure among humans bitten by *Bungarus multicinctus.*

- Kappa-bungarotoxin competes with acetylcholine for the binding site of the cholinergic receptor;
- Permanently binds to the binding site of the cholinergic receptor causes action potentials to be always generated at the post-synaptic membrane;
- Voltage-gated sodium ion channels are always opened and influx of sodium ions will always flux into the axon;
- Depolarization of the post-synaptic neuron will always occur;
- Hence, always triggering the contraction of contractile elements within the muscle fibers and causing respiratory failure as the muscle fibres cannot relax;

[4]

For Examiner's Use

Antivenom is a medication used to treat venomous bites and is recommended for use via injection if the venom from the snake is of high risk of toxicity.

(f) Suggest how antivenom can alleviate the effects of kappa-bungarotoxin.

- It binds to the kappa-bungarotoxin and changes its conformation such that it can no longer bind to the binding site of the cholinergic receptor;
- It is an enzyme that degrades the kappa-bungarotoxin into simpler substances;
- AVP;

......

...**[1]**

[Total: 13]

2010 MJC Prelim 2 Q3
+ 2010 RI Prelim 2 Q5

For Examiner's Use

6. Fig. 6.1 shows a tyrosine kinase receptor. The effect of insulin binding to this complementary receptor is shown in Fig. 6.2. The boxed regions in Fig. 6.1 and Fig. 6.2 are cysteine-rich domains.

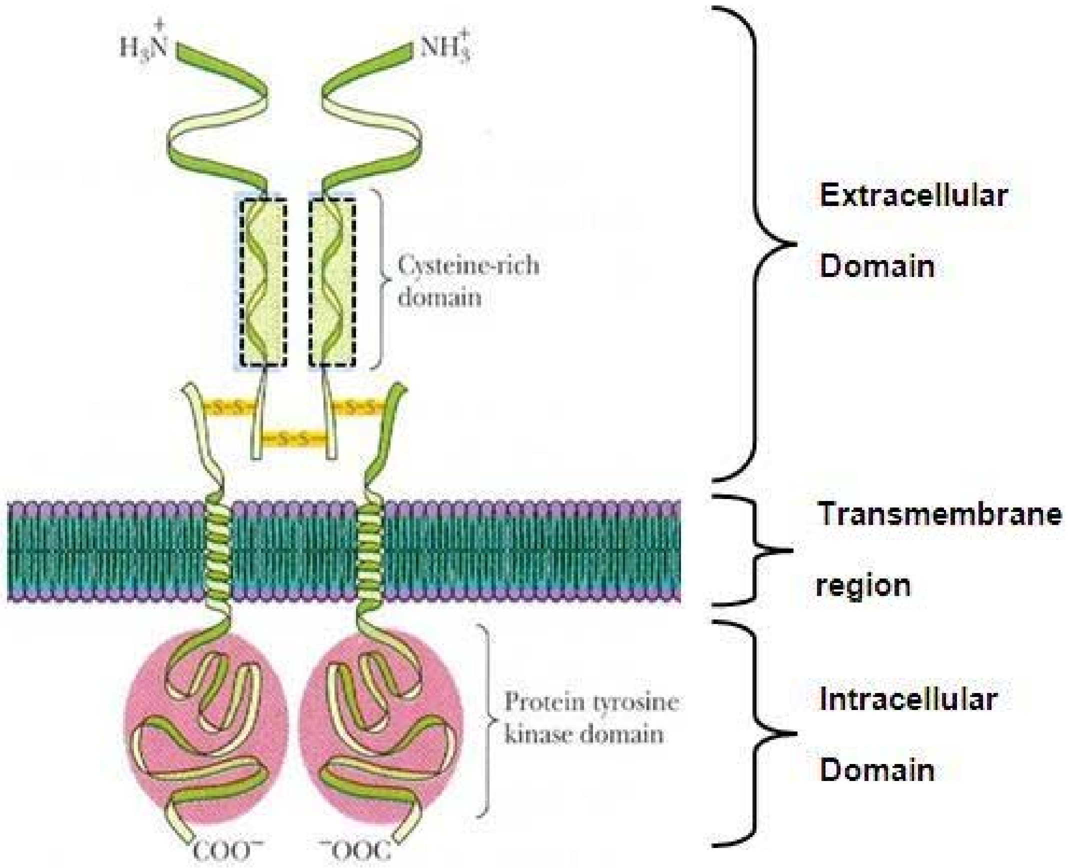
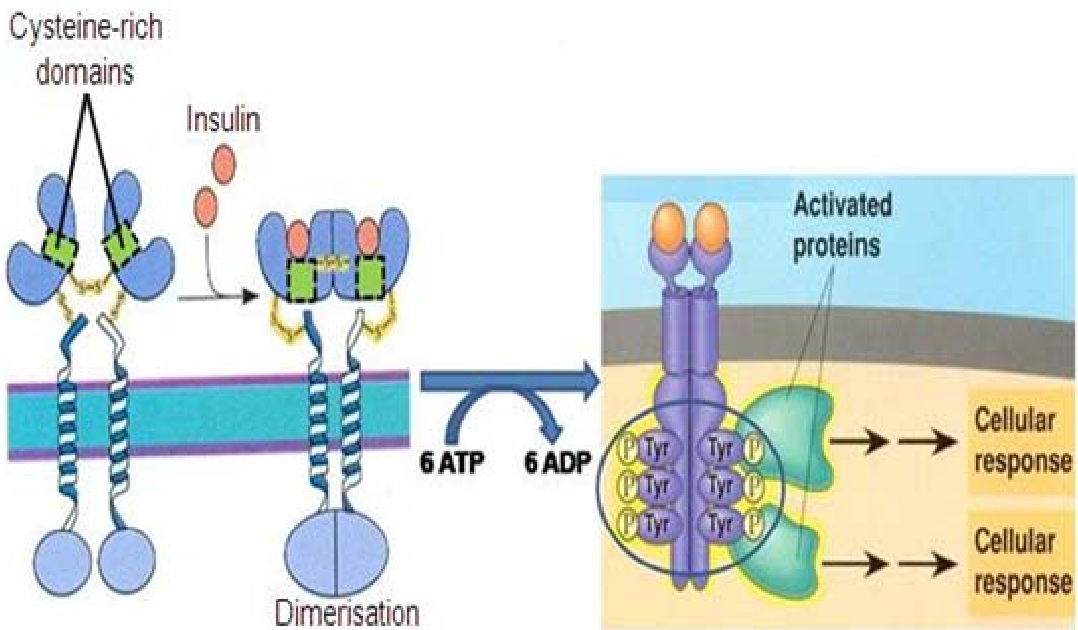


Fig. 6.1



[Turn over

For Examiner's Use

Fig. 6.2

(a) Explain how the structure of the tyrosine kinase receptor is suited for its role in insulin mediated cell signalling.

- Cysteine-rich extracellular domain of receptor is complementary to insulin to facilitate binding;
- Transmembrane helices relays signal to cell interior;
- Dimerisation of intracellular tyrosine kinase domains results in phosphorylation of the tyrosine tails;
- which in turn activate specific relay proteins to elicit cellular responses;

....

....**[2]**

Fig. 6.3 shows how blood levels of glucose and glucagon change after a meal.

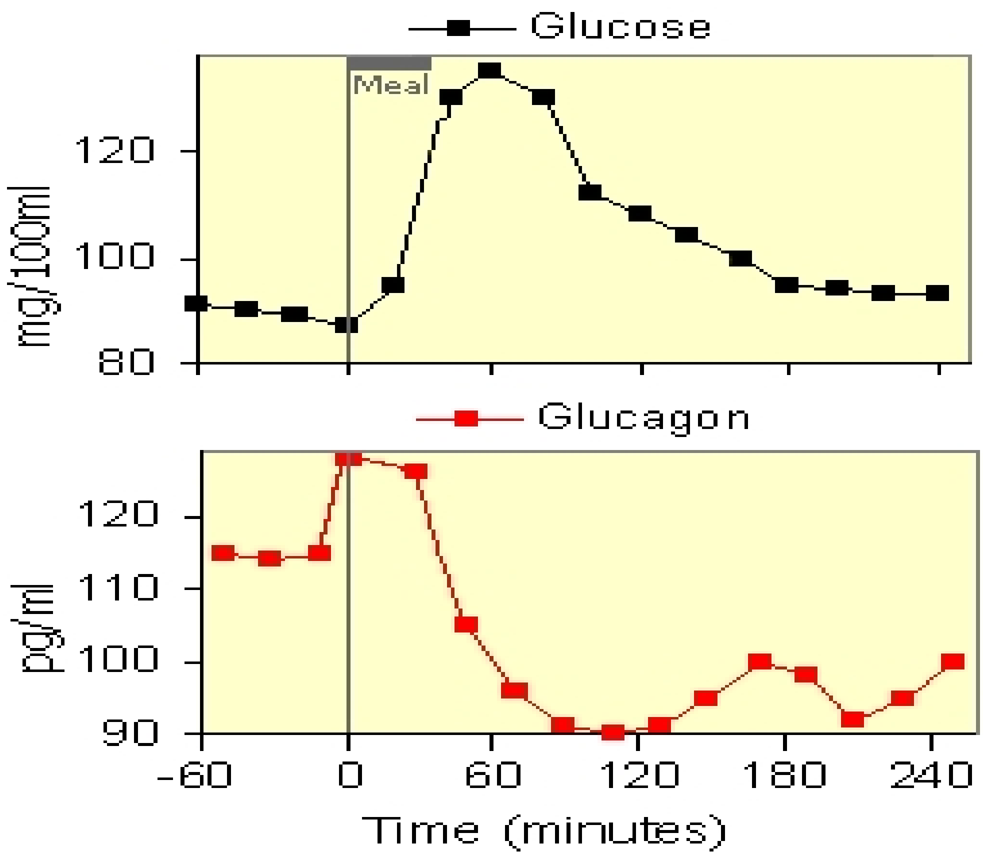


Fig. 6.3

For Examiner's Use

(b) Describe the components of a homeostatic control system and explain the principles of homeostasis.

……………………………………………………………………………………………

- A homeostatic control system would include a receptor / detector to detect stimulus, a control centre to process information from the receptor and effector to carry out the response;
- Self-regulation which refers to the fact that the control mechanism is triggered by the very entity that it serves to regulate.
- Negative feedback - Any deviation from the reference point triggers mechanisms that serve to eliminate that deviation and return to the reference point.

…………………………………………………………………………………………[3]

(c) With reference to Fig. 6.3, explain the relationship between glucose and glucagon levels from 0 to 120 minutes.

- From 0 – 60 mins, glucagon levels decrease from above 120pg/ml to 95pg/ml as glucose levels increase from below 90mg/100ml to above 120mg/100ml;
- From 60 - 120mins, glucose levels gradually decreases from above 120mg/100ml to about 110mg/100ml as glucagon levels decrease from above 120pg/ml to 90pg/ml;

OR

- From 0 – 120 mins, glucagon levels decrease from above 120 pg/ml to 90 pg/ml as glucose levels increases from 85mg/100ml to above 120mg/100ml before dropping to 110mg/100ml; (2 marks)

Max 2 marks

- As glucose levels increase and deviate from set point reference, there is negative feedback due to diminished stimulus to alpha-cells of Islets of Langerhans of pancreas / alpha-cells of Islets of Langerhans of pancreas stops releasing glucagon;
- Decreased secretion of glucagon to blood stream would lead to less glucagon binding to GPCR;
- Conversion of glycogen to glucose (glycogenolysis and gluconeogenesis) is inhibited;

…[4]

Diabetes mellitus is a disease in which high blood glucose cannot be regulated back to normal set point within the body. The hormone insulin is commonly used in the treatment of diabetes. There are two forms of diabetes mellitus: Type 1 and Type 2. Type 2 diabetes mellitus is characterized by insulin resistance whereby the body tissues do not respond effectively to insulin.

(d) Suggest why the body tissue is insensitive to insulin in Type 2 diabetes mellitus.

- The three dimensional configuration/shape of the receptor for insulin may be changed/defective/mutated, hence insulin cannot bind/non-complementary to receptor.
- Receptor-mediated endocytosis of insulin receptors / reduce number of insulin receptors on target cells;

…[1]

For Examiner's Use

Glucagonoma is a rare tumour of the α-cells of the islet of Langerhans which results in an overproduction and secretion of glucagon.

(e) Suggest how glucose metabolism is affected when an individual has a glucagonoma.

- Increase breakdown of glycogen to glucose / glycogenolysis → (high blood glucose levels);
- Increase conversion of amino acids and pyruvate to glucose / gluconeogenesis;
- Increase lipolysis;

…….

….**[1]**

Cancer development occurs in stages. The advanced stage of cancer is characterized by the spread of cancer development to other parts of the body via the circulatory system to form secondary tumours by a process known as metastasis.

(f) Explain the properties of cancer cells required for metastasis.

- Ability to evade white blood cells and prevent them from being degraded;
- No anchorage dependence allows cancer cells to migrate to other parts of the body.
- Angiogenesis/stimulate growth of blood vessels towards itself allows cancer cells to gain access to circulatory system.
- Invasion of surrounding tissue cells allow cancer cells to form secondary tumours in other parts of the body.
- Cancerous cells can adhere to walls of capillary blood vessel;

……………………………………………………………………………………**[2]**

[Total: 13]

2008 ACJC Prelim 2 Q7

For Examiner's Use

7. In New Zealand, there are two species and three sub-species of native bush robins. It is believed that the robins evolved from a common ancestral stock, members of which flew from Australia across the Tasman Sea and became established in New Zealand over a million years ago. This ancestral form is considered to be similar to the present day Australian flame robin, *Petroica multicolor*, a bird with a brightly coloured red breast. The New Zealand birds do not have this red colour. Some characteristics of the birds and their distributions are shown in Fig. 7.1.

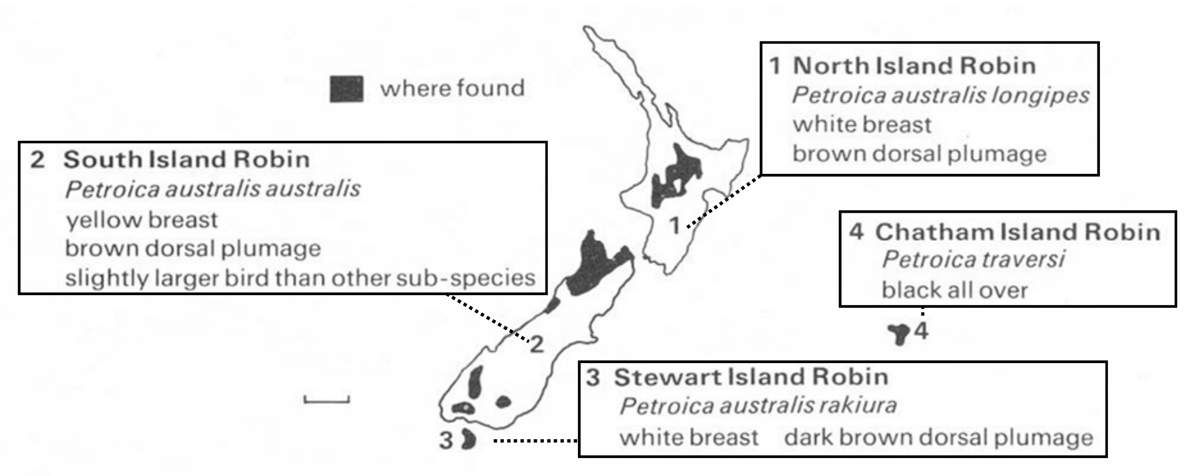


Fig. 7.1

(a) State the scientific name of the two species of native bush robins in New Zealand.

Petroica australis; *Petroica traversi* .... [1]

(b) Explain why the robins in locations **1**, **2** and **3** are similar but different from those in location **4**.

- Birds in locations 1, 2 and 3 are capable of interbreeding (because of the) short distances between the islands;
- (similar environment and so) subjected to similar selection pressure / gene flow is not interrupted / shared a common gene pool for a long time;
- Birds on location 4 could not interbreed with the other birds;
- Allopatric speciation / geographically isolated;
- mutation such as the genes for black plumage would tend to be confined within the isolated inbreeding population;

....[3]

Based on earlier research, the native bush robin species and subspecies were distinguished based on a number of phenotypic differences such as plumage and breast colour. However, with modern technology, researchers have been using molecular methods such as direct DNA and amino acid comparison to further determine the phylogeny between the native bush robin species.

(c) Explain why molecular homology is better than anatomical homology in determining evolutionary relationship between species of native bush robins.

- Similarity in anatomical features could be due to convergent evolution;
- Study of anatomical homology is not possible between morphologically different species;
- Using molecular method, organisms can be compared even if they are morphologically very different + all organisms have certain molecular traits in common, e.g. rRNA sequences or certain fundamental proteins;
- Using morphological features as a benchmark is subjective and non-quantifiable;
- Molecular data is quantifiable and objective. Nucleic acid and amino acid sequence data are precise and easy to quantify, hence allows an objective assessment of evolutionary relationships;
- Fossils obtained from ancestral species may be incomplete thus comparative study of anatomy is not possible;
- Compare molecular divergence of ancestral species with incomplete/no fossil record with that found in other lineages with more complete fossil records;

; for 1 mark, max is 2 marks

....

....

....

...[2]

Differences in the *cytochrome b* DNA sequence of several native bush robins from different regions of New Zealand were measured and plotted against time since divergence from the primitive ancestor as seen in Fig. 7.2.

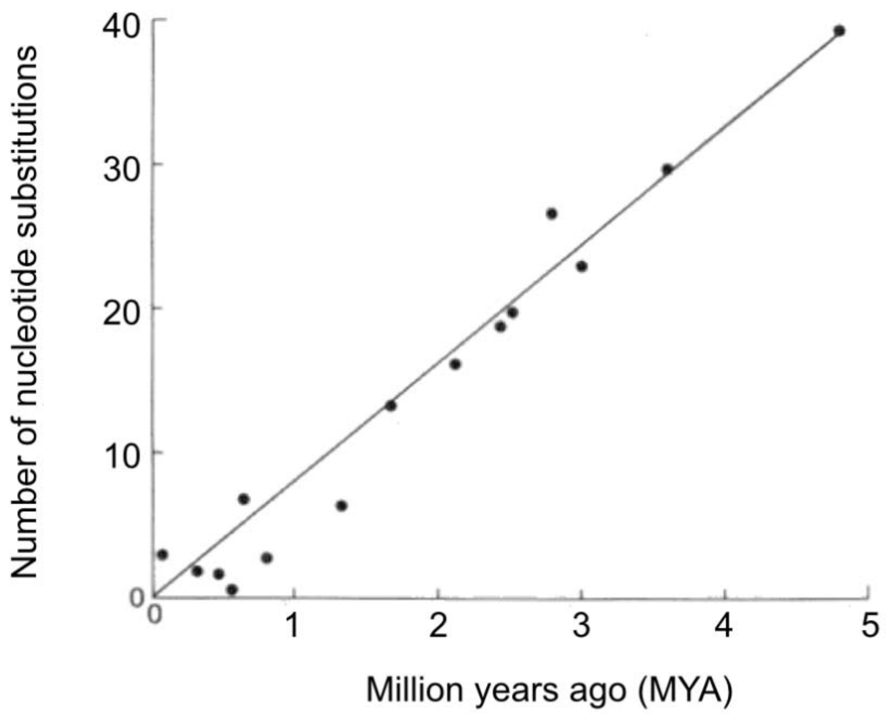


Fig. 7.2

For Examiner's Use

(d) Describe how the differences in the number of nucleotide substitutions support the neutral theory of molecular evolution.

- The plot of the line is straight, indicating that the rate of mutation is constant;
- Changes in the nucleotide sequence arise through neutral mutation;
- for e.g. silent mutation or missense mutation where the change in amino acid does not occur in a critical region of the enzyme;
- There is no effect on the phenotype and fitness of organism and thus allowed to accumulate;
- Small number of changes over millions of years indicating that rate of mutation is slow as
- Cytochrome b gene is a crucial gene in living organisms for cellular respiration;

; for 1 mark, max is 3 marks

[2]

(e) Suggest why New Zealand robins do not have the red breast trait even though it is present in its Australian robin ancestors.

- Australian robins might have gained the red breast by a mutation (which occurred in the Australian population);
- after the colonization of New Zealand by forms which did not possess it / ref to Founder effect;
- Red-breasted forms might have reached New Zealand where the characteristic was at a selective disadvantage / vice versa;
- Frequency of red breast allele decreased and eventually disappeared from the gene pool;
- Ref. to genetic drift such that red breast individuals accidently die;

[1]

[Total: 9]

[Turn over

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a), (b)** etc., as indicated in the question.

8.
(a) Compare between DNA replication and transcription. **[6]**

Similarities
1. Both involve unwinding of the double helix DNA;
2. Both involve breaking of the weak hydrogen bonds between complementary bases;
3. Both involve formation of phosphodiester bonds between neighbouring nucleotides;
4. Both involve aligning free nucleoside triphosphates through complementary base pairing;
5. Both products elongate in 5' to 3' direction;
6. Both occur in the nucleus;

Features	**DNA replication**	**Transcription**
(Main) Enzyme involved in polymerisation ***DO NOT LIST DOWN ALL ENZYMES as it is not a fair comparison. You are afterall comparing processes with different functions! Will you ever ask a child to wrestle with a full grown adult? Common sense must prevail during exams too.***	DNA polymerase - binds to the parental DNA molecule @ the origin of replication	RNA polymerase - binds to the promoter. a region near the beginning of the gene to be transcribed
Enzyme used for unwinding the double helix	Helicase	RNA polymerase
Binding site for start of process	Origin of Replication	Promoter sequence
Raw Material	Deoxyribonucleotides	Ribonucleotides
Template	Both strands of DNA molecule act as a template Whole DNA molecule is being replicated	One strand of DNA molecule acts the template The anti-sense / non-coding / template strand of DNA molecule is being transcribed; the other strand is not transcribed
Base Pairing	Adenine with thymine and vice verse Cytosine with guanine and vice versa	Adenine with uracil and vice verse Cytosine with guanine and vice versa
Presence of proofreading property on enzyme involved	DNA polymerase proofreads the newly synthesized daughter strand, ensuring precise complementary base pairing	NIL

Products	2 DNA molecules. Each DNA molecule comprises of 1 parental strand and 1 complementary daughter strand Products remain in the nucleus	1 RNA molecule Products leave the nucleus
When the process occur	Prior to nuclear division (mitosis and meiosis)	Prior to translation
Purpose	Double the amount of DNA so that: after mitosis - the 2 daughter cells will have the same amt of DNA (2n) as parental cell after meiosis - the 4 daughter cells each will have half (n) the amt of DNA	Protein synthesis

For Examiner's Use

(b) Describe how differences in the structure and organization of prokaryotic and eukaryotic genomes affect their control of gene expression. **[7]**

		Prokaryotic Genome	**Eukaryotic Genome**
Structure	Level of condensation	• DNA not highly condensed / Not bound to histones • Because genome size is smaller	• Many levels of condensation of DNA / DNA bound to histones • Because genome size is larger
	Significance	• No need to decondense chromosome / uncoil DNA from histones before transcription can occur	• Chromatin must be decondensed / have a looser conformation for transcription to occur so that RNA polymerase & transcription factors can bind to DNA
		• Changes in chromosome structure not used as method to regulate transcription	• Rate of transcription can be regulated by histone acetylation • resulting in conversion between euchromatin and heterochromatin, • affecting the ease of transcription
Structure / Organisation	Chromosomes / Genomes bound by membrane	• Not bound by membrane / genome in nucleoid	• Bound by the nuclear membrane / genome in nucleus
	Significance	• Transcription and translation occur simultaneously	• Transcription occurs before translation / transcription occurs within the nucleus and translation outside the nucleus / transcription and translation occur in different locations in the cell
		• Post-transcriptional control of gene expression not possible • Thus mRNA is more easily degraded / less stable	• Post transcriptional control of gene expression is possible • Eg. Alternative splicing / mRNA modifications before translation – 5' cap or poly-A tail • Thus mRNA is less easily degraded / more stable
Structure / Organisation	Presence / Absence of introns	• No introns	• Presence of introns
	Need for mRNA splicing & allows for alternative splicing	• No splicing of mRNA is needed	• Splicing of pre-mRNA is needed / exons / coding regions must be spliced together
		• No alternative splicing possible • Only one mRNA / protein product per gene	• Allows for alternative splicing • giving rise to varying mature mRNA molecules and hence multiple possible protein variants from the same gene
Organisation	Presence / absence of operon	• Genes coding for enzymes with related function grouped in an operon	• Dispersed genes coding for enzymes with related function each controlled by an individual / separate promoter
	Need for coordinate control, eukary genes having	• Produces polycistronic mRNA	• Produces monocistronic mRNA

	shared control elements	• Allows coordinated control of metabolic enzymes involved in the same pathway	• Coordinated control occurs via binding of transcription factors to control elements shared by genes with related functions (when genes are on diff chrom) • Coordinated control also occurs by DNA methylation / histone acetlyation that causes all the genes in the same region to be made available or unavailable for transcription (when genes w related fns are close together on the same chrom)
		• All gene products of an operon are produced in the same amount	• Each gene product can be produced in different amounts or not at all
Organisation	Presence / Absence of (distal) control elements	• No distal control elements or its regulatory sequences are close to promoter	• Presence of distal control elements
	Allows for more fine-tuned control of transcription	• Simple control possible, either transcription switched on or off • Level of expression cannot be finely tuned / regulated as specifically	• Allows for fine tuned control of gene expression / allows for different levels of expression of a particular gene • depending on the activators and repressors present in the cell at a specific time

For Examiner's Use

(c) Outline the viral reproduction cycle of HIV. **[7]**

Stage	**Description**
Attachment / Adsorption	• gp120 glycoproteins on the surface of the HIV binds to CD4 receptors on T helper cells and macrophage. o CD4 (cluster of differentiation 4) is a glycoprotein expressed on the surface of T helper cells, regulatory T cells, monocytes, macrophages, and dendritic cells. • Binding of gp120 to CD4 is mediated by conformational changes in the gp120 protein but such conformational change is not sufficient for fusion. • The co-receptor CCR5 produces a conformational change in gp41 on HIV virion which allows fusion of HIV to host cell membrane.
Penetration	• The viral envelope of HIV fuses with the host cell membrane, releasing the nucleocapsid into the host cell, leaving the viral envelope outside the host cell
Uncoating	• The capsid is degraded, releasing the viral enzymes (integrase, protease and reverse transcriptase) as well as the viral RNA genome into the host cell cytoplasm.
Synthesis / Replication	• HIV's reverse transcriptase uses viral RNA as a template to form a complementary strand of DNA. This forms an RNA-DNA hybrid. • Next, the RNA strand is degraded and reverse transcriptase proceeds to use the remaining DNA strand as a template to form another complementary DNA strand. This forms a dsDNA molecule. • The dsDNA enters the nucleus and is integrated into the host genome using the enzyme integrase and the integrated viral DNA is known as a provirus; • When the provirus is activated, the host RNA polymerase transcribes the viral DNA into viral RNA molecule. • The new RNA genomes serve as nucleic acid for new HIV. • Proviral DNA is also transcribed into viral mRNA, which is then translated by the host cell machinery to produce a single long chain of HIV protein (polyprotein). • Capsid proteins and the various viral enzymes are synthesized by free ribosomes in the cytosol. • Viral glycoproteins (gp 120 and gp 41) are synthesized by fixed ribosomes of the RER and are transported to the GA for chemical modification before incorporation into the host cell membrane. Max 2 marks
Assembly	• Vesicles embedded with viral glycoproteins migrate towards and fuse with the cell surface membrane. • Assembly of two single-stranded RNA molecules associated with reverse transcriptase, integrase and protease occurs within an assembled capsid.
Maturation & Release	• Viral maturation occurs when the polyprotein is cleaved into smaller functional proteins by viral protease. • The virus leaves the host cell via budding, with the host cell membrane forming the new viral envelope.

[Total: 20]

For Examiner's Use

9.
(a) Contrast the structures of viral, prokaryotic and eukaryotic genome. **[5]**

	Viral genome	**Prokaryotic genome**	**Eukaryotic genome**
Genetic material	Either DNA or RNA;	DNA;	DNA;
	Either double-stranded; or single-stranded;	Double-stranded;	Double-stranded;
	Either linear; or circular ;	Circular;	Linear;
	Either segmented; or single;	Singular chromosome	segmented / many different chromosomes;
Presence of telomeres	May or may not be present;	None;	Present
Presence of origin of replication	May or may not be present;	One present	Yes / many present
DNA enclosed in …	capsid;	Nucleoid body	nucleus;
DNA associated with histones	No;	No;	Yes;
Amount of noncoding DNA	Little ;	Little;	Large amounts;
Introns	Absent;	Absent;	Present;
Amount of repetitive DNA / centromeres	Absent	Absent	Large amounts/ Present
spacer DNA between genes	Little	Little	Large amounts
Presence of introns in genes	No (in most cases);	No	Yes;
Size	Smallest;	Smaller	Larger;

1 mark per row. Max 6 marks.

For Examiner's Use

(b) Relate the structure of ribosome to its role in protein synthesis. **[7]**

- Made out of ribosomal RNA and proteins.
- Consists of a small subunit and a large subunit
- Prokaryotes have 70S ribosomes, each consisting of a small (30S) and a large (50S) subunit.

  OR
- Eukaryotes have 80S ribosomes, each consisting of a small (40S) and large (60S) subunit.
- small sub-unit allows attachment of mRNA to form initiation complex
- large sub-unit contain 'P' site: peptidyl-tRNA site, 'A' site: aminoacyl-tRNA site, 'E' site: exit site
- large subunit of the ribosome contains peptidyl transferase, for formation of peptide bond/elongation of polypeptide chain
- A site for receiving the aminoacyl-tRNA complex
- P site holds the 1^{s} aa-tRNA complex/elongating polypeptide chain
- the 'E' site is where the tRNA released into the cytoplasm
- freely floating Ribosomes in the cytosol – produce proteins that function within the cytosol
- Ribosomes attached to the endoplasmic reticulum - synthesise proteins that are meant for insertion into the membrane, for packaging within certain organelles e.g. lysosomes or for secretion out of the cell

For Examiner's Use

(c) Outline the processes involved in oxidative phosphorylation. **[8]**

- Oxidative phosphorylation is the process where ATP is generated using the energy released during transfer of electrons from NADH or $FADH_2$ to oxygen via a series of electron carriers;
- NADH and 2 $FADH_2$ are responsible for donating electrons to the electron transport chain (ETC) during oxidative phosphorylation;
- As electrons move from one carrier to the next, they are moving from a higher energy level to a lower energy level, therefore they lose potential energy;
- At the end of the ETC, electrons combine with the final electron acceptor oxygen and H^+ in the mitochondrial matrix → oxygen is reduced to water;
- This energy is used by ETC carriers to pump H^+ from the matrix into the intermembrane space;
- The pumping of H^+ across the inner mitochondrial membrane against its concentration gradient generates a proton gradient;
- The potential energy contained in this electrochemical gradient is known as the proton-motive force;
- Chemiosmosis is an energy-coupling mechanism that uses the proton-motive force to drive generation of ATP from ADP and $P_{i;}$
- Due to the proton-motive force, H^+ tends to diffuse down its gradient and leak back from the intermembrane space into the matrix;
- H^+ can pass through a channel in ATP synthase and the energy released from the movement of H^+ down its gradient is used to form ATP;
- ATP synthase can use the exergonic flow of H^+ to drive the phosphorylation of ATP from ADP and P_i.;
- 34 ATP molecules formed from 1 glucose molecule after oxidative phosphorylation;

End of Paper